

Volume I. Introduction.

We presuppose, in our considerations, the natural numbers $1, 2, 3, \dots$ and the rules according to which calculations are made with these numbers as known and given. The fundamental arithmetical operations, the so-called four species, are addition, multiplication, to which, as repetition, exponentiation belongs, subtraction, and division. The first two are called the direct operations of calculation; they are distinguished by the fact that in the realm of the natural numbers they can be carried out without restriction. The first of the indirect or inverse operations, subtraction, can be carried out only when the minuend is greater than the subtrahend.

The problem of division can be understood in two ways. In the first elementary interpretation one asks how often the divisor is contained in the dividend. A number is not contained in a smaller one. But if the dividend is equal to or greater than the divisor, then the answer to the question gives a quotient and, in most cases, a remainder which is smaller than the divisor. If no remainder is left, one says that the division comes out exactly, or that the dividend is divisible by the divisor, or that the divisor is a factor or divisor of the number which forms the dividend.

This problem, which already presents itself at the first stages of the art of calculation, leads to a more deeply lying distinction among numbers, which is the foundation of all number theory.

Since a factor can never be greater than the number of which it is a factor, every number has only a finite number of factors. Every number is divisible by 1 and by itself, and a number which has no other divisor is called a prime number. The number 1 itself is customarily not designated a prime number, for reasons of expediency. If two numbers are divisible by a third, then the sum and the difference of the first two are also divisible by the third; and if one number is divisible by a second and this second by a third, then the first is also divisible by the third.

Two numbers always have the common divisor 1. If they have no other common divisor, as for example the pairs of numbers 5 and 7, or 21 and 38, then the two numbers are called relative prime numbers, or also numbers prime to one another.

Among the common divisors of any two given numbers one will be the greatest, and it is a very important problem to find this greatest common divisor. This is accomplished by a procedure known under the name algorithm of the greatest common divisor, which already occurs in Euclid.¹

Let a, a_1 be the two numbers; take the larger of them, say a , as dividend and the smaller a_1 as divisor, and determine the quotient q_1 and, if the division does not come out exactly, the remainder a_2 , so that

$$a = q_1 a_1 + a_2,$$

and $a_2 < a_1$. Every common divisor of a and a_1 is then also a common divisor of a_1 and a_2 , and conversely. If one proceeds with a_1, a_2 just as with a, a_1 , and puts, when the division does not come out exactly,

$$a_1 = q_2 a_2 + a_3,$$

then again $a_3 < a_2$, and every common divisor of a_1 and a_2 is also a common divisor of a_2 and a_3 , and conversely. If one continues dividing in this way, then, since the numbers $a, a_1, a_2, a_3, \dots$ continually decrease, the division must necessarily come out exactly after a finite number of steps; that is, at last a pair of equations

$$a_{\nu-2} = q_{\nu-1} a_{\nu-1} + a_{\nu}, \quad a_{\nu-1} = q_{\nu} a_{\nu}$$

must occur. Then a_{ν} is a common divisor of all preceding a 's, hence also of a and a_1 , and every common divisor of a and a_1 is a divisor of a_{ν} . Thus a_{ν} is the greatest common divisor of a and a_1 , and at the same time we have arrived at the theorem that every common divisor of two numbers must divide their greatest common divisor.

¹Elements, Book VII, 11, vol. II of Heiberg's edition.

If a and a_1 are relatively prime numbers, then the last divisor is $a_\nu = 1$. Multiplying, under this assumption, the preceding equations by an arbitrary number b , it follows that b is the greatest common divisor of ab and a_1b , and hence that every common divisor of ab and a_1 , or of a and a_1b , must be a divisor of b . Thus if a_1 is relatively prime to a and to b , it is also relatively prime to ab ; and from the special assumption that a_1 is a prime number it follows that a product can be divisible by a prime number only when at least one of the factors is divisible by it. Thus, if the product ab is divisible by a_1 and a_1 is relatively prime to a , then b must be divisible by a_1 .

Let a, b be any two numbers with greatest common divisor d , and let $a = da'$, $b = db'$. Then a' and b' are relatively prime to one another. Every number m which is at once a multiple of a and of b has the form $m = am' = da'm'$, and in this $a'm'$ must be a multiple of b' ; therefore m' too must be a multiple of b' . That is, every number which is at once a multiple of a and of b is divisible by $a'b'd$. This number $a'b'd$, which itself is a common multiple of a and b , is therefore called the least common multiple of a and b .

If a number is divisible by two relatively prime numbers, then it is also divisible by their product; and if, therefore, a number m is divisible by several numbers of which each two are relatively prime to one another, then it is also divisible by the product of all these numbers.

On this rests the proof of the important theorem that a number m can always be represented, and in only one way, as a product of prime numbers. For since a divisor cannot be greater than the dividend, m can certainly be divisible by only a finite number of prime numbers. If a is one of these prime numbers and a^α the highest power of a which divides m , then α too is a definite number. Let likewise $b^\beta, c^\gamma, \dots$ be the highest powers of the remaining prime numbers $b, c, \dots$ which divide m . Then, since any two of the numbers $a^\alpha, b^\beta, c^\gamma, \dots$ are relatively prime, m must also be divisible by the product $a^\alpha b^\beta c^\gamma \dots$; and it must be equal to this product, since otherwise another prime number, or a higher power of one of the prime numbers $a, b, c, \dots$, would have to divide m .

We now give an overview of the extensions of the concept of number which are necessary in mathematics and which were gradually introduced.

By a manifold or set, or by the abbreviated sign M , we understand a system of objects or elements of any kind, which is so delimited and complete in itself that, for every arbitrary object, it is completely determined whether it belongs to the system or not, regardless of whether we are able, in each particular case, actually to make the decision or not.

A set is called ordered when, of any two different elements, one that is fully determined in itself always counts as the greater, and in such a way that from $a > b$, $b > c$ one always has $a > c$. If $a > b$ and $b > c$, or $a < b$ and $b < c$, we say that b lies between a and c .

The natural numbers form an ordered set; between two successive elements of it there lies no further element. Such a manifold is called discrete. An ordered set with the property that between any two elements further elements can always be found is called dense. One can form a dense set by combining the natural numbers in pairs and regarding these pairs as elements of a set. These pairs shall be called fractions and denoted by $m : n$ or m/n , and two such fractions $m : n$ and $m' : n'$ are set equal when $mn' = nm'$. If all fractions equal to one another are combined into one element, one obtains a manifold which is ordered once one further stipulates that $m : n$ is greater than $m' : n'$ when $mn' > nm'$. That this manifold is dense is seen as follows: if $\mu = m : n$, $\mu' = m' : n'$ are two fractions and $\mu > \mu'$, then, with h arbitrary, one can put

$$\frac{hmn'}{hnn'}, \quad \frac{hm'n}{hnn'},$$

and therein $hmn' > hm'n$. One can always take h so large that further numbers lie between hmn' and $hm'n$, and if p is such a number, then $p : hnn'$ lies between μ and μ' .

The points of a straight line can also be regarded as an ordered set if by greater and smaller one understands some indication of position, for instance farther right and farther left, or higher and lower.

A division of an ordered set M into two parts A, B of such a kind that every element a of A is smaller than every element b of B is called a cut in M and is suitably denoted by (A, B) . Such a

cut arises when one selects an element μ of M , assigns all smaller elements to A , all larger elements to B , and assigns μ itself at will to A or to B . Strictly speaking, according as one does the one or the other, two cuts arise, but we shall always regard them as equal. If, in a cut (A, B) , either A contains a greatest element or B a least element μ , then we say that μ produces the cut (A, B) . But it may also occur that neither A has a greatest nor B a least element.

If every cut in a dense set is produced by a definite element μ , then the set is called continuous.

Continuity as well as density are properties which, by the nature of the matter, are inaccessible to our sense perception; consequently they also cannot be demonstrated with rigor in things of the external world, in spatial magnitudes, intervals of time, or masses, however much they may seem to lie in the essence of our intuition. But pure conceptual systems can very well be constructed to which density without continuity, or also density and continuity, belong.²

This very abstract way of looking at the matter gives us the assurance that the assumption of a continuous set contains no contradiction, and that such sets at least exist in the realm of thought. Geometry, and also analysis, which has always willingly attached itself to geometrical intuition, long tacitly assumed the existence of continuous sets, for example in the points of a straight line or of any other connected line-segment, as a kind of axiom. The difference between a dense and a continuous set, which lies at the basis of the distinction between commensurable and incommensurable segments, was also not missed by the ancients.³

We too shall in what follows not renounce the aid of geometrical intuition, and shall, for example, without hesitation regard the points of a straight line as a continuous set.

An ordered set M is called measurable under the following assumptions. Addition and multiplication are generally executable in M , and likewise the subtraction of a smaller element from a greater one; that is, from any two elements a, b (which may also be identical) a new element $a + b$ of M can be derived according to a definite rule, in such a way that $a + b$ is greater than a and greater than b , and the familiar laws of addition expressed in the formulae

$$a + b = b + a, \quad (a + b) + c = a + (b + c)$$

hold. Moreover, for two elements a, c , of which the second is the greater, a third element b can be found such that $a + b = c$, which is also expressed by the sign of subtraction $b = c - a$. Thus two unequal elements of M always have a definite difference. From these assumptions it follows that a sum becomes greater when one of its summands becomes greater. The repeated, say m -fold, addition of the same element a is called multiplication, and its result is denoted by ma .

Finally there is added one more assumption, namely that, for every given a , a sufficiently high multiple ma is greater than any other element b given in advance. Among the elements of a measurable set there is therefore no greatest element.

In a dense measurable set there is also no least element. For if a were the least and b an arbitrary element, then between b and $b + a$ no elements could lie; for if $b < c < b + a$, then it would follow from the definition of measurability that $a' = c - b$ would be smaller than a . Conversely it also follows that a measurable set in which no least element occurs is dense. For if a and $a + b$ are any two elements, one need only add to a an element smaller than b in order to obtain an element between a and $a + b$.

If a set is continuous, then the assumptions for measurability can be simplified still further, because then subtraction is a consequence of addition. For if a and c are two elements of a continuous ordered set M in which addition exists, and if $c > a$, then one obtains a cut (A, B) in M by assigning to A all elements x for which $a + x < c$, and to B those for which $a + x > c$. This cut is produced by an element b , for which $a + b = c$ must hold.

²This was proved by Dedekind, to whom we owe in general the definition of continuity given above. Compare Dedekind's writings, *Stetigkeit und irrationale Zahlen*, Braunschweig 1872, 1892, and *Was sind und was sollen die Zahlen?*, Braunschweig 1888, 1893. Other manifolds to which continuity belongs were formed by Weierstrass and G. Cantor.

³Euclid, Elements, Book X.

The natural numbers form, according to our definition, a measurable set in which a least element, namely 1, occurs. The manifold of rational fractions is likewise measurable when addition and subtraction are explained according to the known rules of fractional calculation. Especially important, and as it were typical for the measurable sets, is the manifold of rectilinear segments, or lengths of a line, which are added simply by laying them end to end. Quantities of material, compared by means of the balance, and intervals of time, measured by the clock, also furnish examples of measurable sets. The manner of measuring does not lie in the nature of the manifold itself, but is put into it by the thinking observer; thus, for example, it would be just as admissible to understand by the sum $a + b$ of two segments a and b the hypotenuse of a right triangle with legs a, b , instead of, as is usually assumed, the segment composed from a and b by laying them end to end.

In order to make the continuous set B of cuts in the manifold R of rational fractions into a measurable one, observe first the following. If $\alpha = (A, B)$ is an element in B and μ is an arbitrarily given rational fraction, then one can always choose an element a' in A so that $a' + \mu = b'$ is contained in B . For if one chooses two arbitrary elements a, b , then one can determine the natural number m so that $m\mu > b - a$, hence $a + m\mu$ is contained in B . If then h is the smallest integer for which $a + h\mu$ is contained in B , then $a + (h - 1)\mu = a'$ is in A and hence $a' + \mu = b'$ is in B .

We now understand by the sum $(A, B) + (A', B') = (A'', B'')$, or $\alpha + \alpha' = \alpha''$, the cut in R which is obtained by assigning a rational fraction a'' to A'' only when an a in A and an a' in A' exist such that $a'' \leq a + a'$. In fact (A'', B'') is a cut; for if a'' is contained in A'' , the same holds for every smaller fraction, and there are fractions which are in A'' and fractions which are not in A'' , namely fractions of the forms $a + a'$ and $b + b'$. Further, α'' is greater than α and than α' . For A'' first contains all of A , and if a' is an arbitrary fraction in A' , one can choose a in A so that the element $a + a'$ of A' is contained in B ; hence A'' is more comprehensive than A . The cuts produced by rational fractions μ, μ' give, by addition, the cut produced by $\mu + \mu'$.

We now pass to the definition of ratios, which since antiquity have been regarded as the foundation of the theory of numbers, and here follow Euclid first of all.⁴

If one combines the elements of a measurable set M into pairs, and regards these pairs themselves as elements, then a new manifold arises. We denote such a pair by $a : b$, or also by a/b ; but, when a and b are different elements, we distinguish $a : b$ from $b : a$, and call a the numerator and b the denominator of $a : b$. We shall call these pairs ratios, and shall now order and make measurable this new set.

Assume first that two whole numbers m, n exist such that $na = mb$, as is always the case, for example, when M is the system of natural numbers, or when a, b are two commensurable segments. Then, if p, q are two other whole numbers, $qa = pb$ holds if and only if $mq = np$. The pair of numbers p, q is completely determined by this requirement if the condition is added that p, q are to be relatively prime. Then, if h is an arbitrary whole number, $m = hp, n = hq$. In this case we call the ratio $a : b$ rational and set it equal to the rational fraction $m : n$ or $p : q$. Accordingly these rational fractions may be regarded as ratios of whole numbers.

All rational ratios equal to one another form a rational number, and the rational numbers form, like the rational fractions, an ordered, dense, and measurable manifold. The natural numbers themselves are inserted into the manifold of rational numbers if by the natural number m one understands the ratio $m : 1$.

We now return to any measurable manifold M and take from it any two elements a and b . If one chooses, which is always possible, two natural numbers m, n so that $na > mb$, then the ratio $a : b$ is called greater than the rational ratio $m : n$, or

$$\frac{a}{b} > \frac{m}{n};$$

⁴Elements, Book V.

and if $m : n = p : q$, then also $a : b > p : q$. Similarly, if $n'a < m'b$, then

$$\frac{a}{b} < \frac{m'}{n'}.$$

If $a : b > m : n$, then one can insert one, and consequently also arbitrarily many, rational numbers $m_1 : n_1$ such that

$$\frac{a}{b} > \frac{m_1}{n_1} > \frac{m}{n}.$$

For this choose an arbitrary whole number k and determine the whole number h so that $h(na - mb) > kb$, which is always possible. Then

$$\frac{a}{b} > \frac{hm + k}{hn} > \frac{m}{n}.$$

And in exactly the same way, if $n'a < m'b$ and $h'(m'b - n'a) > k'a$, one obtains

$$\frac{a}{b} < \frac{h'm'}{h'n' + k'} < \frac{m'}{n'}.$$

If now $a : b$ and $\alpha : \beta$ are any two ratios, which for brevity we shall also denote by ϵ, ϵ' , and whose elements belong to the same or to different manifolds, then two cases are possible: either no rational ratio μ lies between ϵ and ϵ' , or a rational ratio lies between them.

In the first case the two ratios ϵ and ϵ' are called equal. One sees that two ratios which are equal to a third are also equal to one another; for if $\epsilon < \mu < \epsilon'$, then every other ratio ϵ'' is either smaller than, equal to, or greater than μ . If ϵ'' is equal to μ , then rational ratios lie both between ϵ and ϵ'' and between ϵ'' and ϵ' . If, however, ϵ'' is smaller than μ , then μ lies between ϵ'' and ϵ' , and if ϵ'' is greater than μ , then μ lies between ϵ and ϵ'' ; hence ϵ'' cannot at the same time be equal to ϵ and equal to ϵ' .

In the second case the ratios ϵ, ϵ' are called unequal. Then either $\epsilon < \mu < \epsilon'$ or $\epsilon > \mu' > \epsilon'$ must hold, and these two cases exclude one another; for from $\epsilon < \mu < \epsilon'$ and $\epsilon > \mu' > \epsilon'$ it would follow that $\epsilon < \mu'$ and consequently $\mu < \mu'$. The relation of magnitude remains unchanged when ϵ or ϵ' is replaced by an equal element. In the case $\epsilon < \mu < \epsilon'$ the ratio ϵ is called smaller than ϵ' , and in the opposite case greater. Between two unequal ratios one can insert an arbitrary number of rational ratios.

If now we collect all ratios equal to one another, we obtain a generic concept which we designate as a number in the general sense of the word. A number is therefore a name or sign for a certain manifold whose elements are precisely the ratios equal to one of them.⁵ Under this concept of number are included the rational ratios, and hence also the natural numbers as the ratios $m : 1$; they form the rational numbers. Numbers which do not arise from rational ratios are called irrational numbers. According to what has been developed so far, the numbers form an ordered set, and their order can be fixed if, for each number, any one of the ratios contained under it is chosen as representative.

Of two ratios with the same denominator and unequal numerators, that one is greater whose numerator is greater; and of two ratios with the same numerator and unequal denominators, that one is smaller whose denominator is greater. For if a, a', b are arbitrary elements of a measurable set and $a' > a$, choose first a whole number n such that $na > b$ and $n(a' - a) > b$. Then take m to be the least whole number satisfying $mb > na$; then $na < mb$, but $mb < na'$. Thus

$$\frac{a}{b} < \frac{m}{n} < \frac{a'}{b},$$

and hence $a : b < a' : b$. In exactly the same way one proves that if $b' > b$, then $a : b > a : b'$. This theorem is also expressed by saying that a ratio increases with the numerator and decreases with increasing denominator.

⁵The natural numbers too can be traced back in a simple and consistent way to the generic concept.

From this one easily obtains the following theorem. If a, b, c, d are elements of the same measurable set and $a : b = c : d$, then also $a : c = b : d$. For suppose that $a : c < b : d$; then there would have to be two whole numbers m, n such that

$$na < mc, \quad nb > md.$$

Then, however, by the theorem just proved, $na : nb < mc : md$, and hence $a : b < c : d$, contrary to the supposition.

To this is now attached the following principal theorem. If of four quantities a, b, c, d any three are given arbitrarily from a continuous measurable set, then the fourth can be determined in the same set so that $a : b = c : d$.

The theorem is an immediate consequence of the assumed continuity. For if, for example, an element x of M is put into A or into B according as $x : b$ is smaller or greater than $c : d$, then a cut is obtained, which is produced by an element a satisfying $a : b = c : d$. But this theorem also holds in certain non-continuous sets, for example for the rational fractions.

It follows that as representatives of two numbers one can always choose two ratios whose elements belong to the same manifold and have the same arbitrarily chosen denominator. The addition is then defined by

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}.$$

This rule contains as a special case the addition of rational fractions; and to justify it generally it remains only to show that if $a : c = a' : c'$ and $b : c = b' : c'$, then also $(a+b) : c = (a'+b') : c'$. We prove this by showing that if $a : c = a' : c'$ and $(a+b) : c > (a'+b') : c'$, then also $b : c > b' : c'$ must hold. Let therefore, with m, n two whole numbers,

$$\frac{a+b}{c} > \frac{m}{n} > \frac{a'+b'}{c'}.$$

Then $n(a+b) > mc$ and hence

$$\frac{b}{c} > \frac{mc - na}{nc}.$$

On the other hand, from the supposition $a : c = a' : c'$ it follows easily that

$$\frac{mc - na}{nc} = \frac{mc' - na'}{nc'},$$

and therefore $b : c > b' : c'$, as was required to be proved.

It is thus shown that the numbers, as we have defined them, also form a measurable set. If a and c are taken from a continuous manifold, then, with fixed c , the ratios $a : c$ also form a continuous set; hence, since continuous sets exist at all, it follows that the numbers too form a continuous set.

If $\alpha, \beta, \gamma, \delta$ are now numbers, then from the proportion $\alpha : \beta = \gamma : \delta$ any one of the four numbers can be determined by the three others given. If one puts $\delta = 1$ and seeks α , one obtains the multiplication $\alpha = \beta\gamma$; the commutativity of the factors is a consequence of the theorem that from $\alpha : \beta = \gamma : \delta$ there follows $\alpha : \gamma = \beta : \delta$. If one seeks γ , one obtains division; and from the definition of addition given above there follows the fundamental formula $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$. The four fundamental operations are therefore executable in the domain of numbers, with the sole restriction that in subtraction the subtrahend must be smaller than the minuend.

The construction of a cut in the sequence of numbers always supplies the proof for the existence of a number satisfying prescribed requirements. Thus one obtains a cut (A, B) by putting into A all and only those numbers whose square is smaller than a prescribed number α ; to this cut there corresponds a definite number whose square is equal to α and which is denoted by $\sqrt{\alpha}$. In this way the existence of square roots is proved.

The numerical sequences introduced by G. Cantor for the definition of irrational numbers can also be referred back to cuts.⁶

According to Cantor, by a numerical sequence one is to understand any unbounded system of numbers ordered in a definite way

$$S = x_1, x_2, x_3, x_4, \dots$$

having the property that there is a definite number g below which none of the numbers of S sinks, and that, if δ is any chosen number and x_n, x_m are different elements of S , then $x_n - x_m$ or $x_m - x_n$ always remains smaller than δ when m and n have exceeded a sufficiently large number.

There are always numbers which are not exceeded by the numbers of such a sequence; for, once δ has been chosen and n suitably determined, x_m , whatever the value of m , never exceeds the greatest of the numbers $x_1, x_2, \dots, x_n, x_n + \delta$. One now obtains a cut (A, B) by throwing into B those numbers which, when n has a sufficiently high value, are no longer exceeded by any x_n , and all other numbers, which are therefore exceeded by infinitely many x_n , into A . If this cut is produced by the number α , then, however small ε may be, there are always infinitely many numbers x_n between $\alpha - \varepsilon$ and α ; and one may say that this number α , completely determined by S , is produced by the numerical sequence S . According to Cantor the numerical sequence S is directly the definition of the number α . Of course one and the same number α can be produced by very different numerical sequences. These numerical sequences, however, are all to be regarded as equal to one another. Among other things, one may take all numbers of S to be rational fractions. Conversely, it is also easily shown that for every given number α one can always give numerical sequences S by which α is produced, so that the totality of the numerical sequences S likewise forms a continuous set.

In the explanation of the fundamental operations an inconvenient restriction has appeared in subtraction, from which we can free ourselves by introducing negative numbers.

Let x denote any element of the number system defined so far, which we shall now call the system of positive numbers. We take this number system a second time and, to distinguish it, denote in this second system, which is to be called the system of negative numbers, every element by $-x$. This second system we now order in exactly the reverse way from the first, so that wherever in the system x "greater" stands, in the system $-x$ "smaller" is put, and conversely. Addition and subtraction in $-x$ are explained just as in x , so that

$$(-x) + (-y) = -(x + y), \quad (-x) - (-y) = -(x - y)$$

shall hold.

But we wish to order these two number systems together in such a way that every $-x$ is to be smaller than every x . We thus obtain an ordered set in which there is no greatest and no least element. This system is also in general continuous; only the single cut $(-x, x)$ is produced by no element, and here, where the two systems meet, a violation of continuity is still present. To restore continuity we must therefore add, corresponding to the cut $(-x, x)$, a number zero, or 0 , which is precisely defined by this cut. Then we have an ordered continuous set unbounded on both sides, the complete series of real numbers.

In this enlarged number domain we now explain addition generally by putting, by definition,

$$x + (-x) = 0, \quad x + 0 = x,$$

$$x + (-y) = x - y \quad \text{if } x > y, \quad x + (-y) = -(y - x) \quad \text{if } y > x.$$

With this explanation of addition, if z_1, z_2, z_3 are any three numbers of the whole number domain, the laws

$$z_1 + z_2 = z_2 + z_1, \quad (z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

⁶Cantor, *Ueber die Ausdehnung eines Satzes aus der Theorie der trigonometrischen Reihen*. Mathematische Annalen, vol. 5 (1872); compare also Heine, *Elemente der Functionenlehre*. Journal fuer Mathematik, vol. 74 (1872).

hold, which are called the commutative and the associative laws. One forms the sum of an arbitrary number of summands by uniting, at will and successively, two summands at a time into a sum. Subtraction no longer needs special consideration if one explains $z_1 - z_2$ by $z_1 + (-z_2)$ and sets $-(-z) = z$.

The number line is represented intuitively by points, by laying off from a fixed point of a straight line, denoted by 0, the positive numbers as segments on one side, say the right, and the negative numbers on the other, the left, side. The image of the sum of two segments $z_1 + z_2$ is obtained by laying off from the point z_1 the segment of length $+z_2$ to the right or to the left, according as z_2 is positive or negative.

Multiplication and division are introduced into the enlarged number domain by the equations

$$x(-y) = (-x)y = -xy, \quad (-x)(-y) = xy, \quad 0x = 0.$$

Division, as the inverse of multiplication, is always possible except when the divisor is zero.

A further extension of the concept of number consists in the introduction of complex quantities. We combine every two numbers of the complete number line into pairs (x, y) , and regard two such pairs (x, y) and (a, b) as equal only when $x = a$, $y = b$. These pairs of numbers form a manifold whose elements cannot indeed be ordered, but with which the arithmetical operations of addition, subtraction, multiplication, and division are to be performed according to the following rules:

$$(x, y) + (a, b) = (x + a, y + b),$$

$$(x, y)(a, b) = (xa - yb, xb + ya).$$

We also stipulate that $(x, 0) = x$, which does not contradict these equations. The pair (x, y) is equal to 0 only when x and y are both zero. For abbreviation we denote $(0, 1)$ by i . The equations above then give the consequences

$$(x, 0)(0, 1) = (0, x) = ix, \quad (x, 0) + (0, y) = (x, y) = x + yi,$$

$$x + yi + a + bi = (x + a) + (y + b)i,$$

and the inverse of addition

$$x + yi - (a + bi) = (x - a) + i(y - b),$$

and further multiplication

$$(x + yi)(a + bi) = xa - yb + i(xb + ya), \quad i^2 = -1,$$

$$(x + yi)(x - yi) = x^2 + y^2,$$

$$\frac{x + yi}{a + bi} = \frac{ax + by + i(ay - bx)}{a^2 + b^2},$$

by which division is defined, except when $a + bi = 0$.

Numbers of the form ix are called purely imaginary numbers, and i the imaginary unit. The numbers $a + bi$ are called imaginary or complex. The systems of real and of purely imaginary numbers are contained among them as special cases. Thus, in the domain of complex numbers $x + yi$, the fundamental operations can be carried out without restriction, except for division by zero; and calculation with real numbers is a special case.

Following Gauss, one represents the complex numbers $z = x + yi$ geometrically by the points of a plane, laying down a rectangular system of coordinates and regarding the point with coordinates x, y as the image of the numerical value z . The points of the x -axis represent the real numbers x in the way discussed above. The points of the y -axis are the images of the purely imaginary numbers yi . The origin of coordinates is the image of the number 0. The radius vector from the zero point

to the point z has the numerical value $\rho = \sqrt{x^2 + y^2}$, which is called the absolute value, or the magnitude, or, according to the older terminology, the modulus of the complex number z .

The only number having absolute value 0 is 0. Every positive number occurs as the absolute value of infinitely many complex numbers, and the image-points of all numbers with the same absolute value lie on a circle whose center is at the origin of coordinates.

Two imaginary numbers which differ only by the sign of i , thus $x + yi$ and $x - yi$, are called conjugate imaginary. Their product is the square of the absolute value of either of them. If one of two conjugate imaginary numbers is zero, then the other is also zero; hence in every correct numerical equation one may replace i by $-i$ without disturbing its correctness.

We shall also record the often used theorem that the absolute value of the sum of two non-zero complex numbers is never greater than the sum of the absolute values of the summands, and is equal to it only when the ratio, or quotient, of the two summands is real and positive. Namely, let

$$\begin{aligned} z &= x + yi, & c &= a + bi, & Z &= z + c = (x + a) + (y + b)i, \\ \rho^2 &= x^2 + y^2, & r^2 &= a^2 + b^2, & R^2 &= (x + a)^2 + (y + b)^2. \end{aligned}$$

Then

$$(\rho + r - R)(\rho + r + R) = (\rho + r)^2 - R^2 = 2(\rho r - ax - by),$$

which is surely positive when $ax + by < \rho r$. If however $ax + by > \rho r$, then it would follow that

$$\rho^2 r^2 - (ax + by)^2 = (ay - bx)^2 < 0,$$

which is impossible; and equality occurs only when $ay - bx = 0$. Thus $\rho + r \geq R$, and only in the special case $\rho + r = R$ when $ay - bx = 0$ and $ax + by > 0$. This is the assertion.

In the geometric representation this theorem says that in a triangle one side is smaller than the sum of the two others. The theorem has the further consequence that the absolute value of a sum cannot be smaller than the difference of the absolute values of the summands. For applying the preceding theorem to the sum $c = Z - z$, one obtains $r \leq R + \rho$, or

$$R \geq r - \rho.$$

Equality here occurs only when the quotient $z : c$ is real and negative.

It remains to say a word about the most important aid of algebra, literal calculation. The use of this aid is so general that one sometimes uses the word literal calculation outright as synonymous with algebra. We presuppose as known the rules according to which calculations are made with such literal expressions. Equations between literal expressions can be of two kinds. Either they are so-called identities, that is, the two expressions set equal to one another can, by applying the rules of calculation, be transformed so that both expressions agree exactly. From such literal equations one then obtains correct numerical equations when the letters are replaced by arbitrary real or complex numbers, provided that the demand to divide by zero does not arise. The letters in such equations are often also designated as variables, because one may imagine, without ever coming to a contradiction, that other and again other numerical values are successively put for the letters.

Another kind of equations between literal expressions does not have this character of identity. They contain rather a requirement imposed on the numbers which may be substituted for the letters without committing an error. Algebra has the task of determining numerical values which satisfy such a requirement, that is, of solving the equation. In these equations the letters are also designated as "unknowns." Very often in one and the same equation letters of two kinds occur: some for which arbitrary numerical values are to be put, and others whose numerical value is first to be determined.

First Section. Rational Functions.

§ 1. Whole functions.

The next object of consideration is whole rational functions, or, briefly, whole functions of one variable. By these we understand expressions of the following form:

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n, \quad (1)$$

where n is an integral exponent, the degree of the function $f(x)$. The degree is a natural number. Sometimes, however, it is also useful to speak of whole rational functions of degree 0, by which a constant is understood. x is called the variable, $a_0, a_1, a_2, \dots, a_{n-1}, a_n$ the coefficients. Both x and $a_0, a_1, \dots, a_n$ are symbols for indeterminate quantities, with which one proceeds according to the rules of literal calculation, and for which definite numerical values may also be substituted under some circumstances (cf. the Introduction).

If the function $f(x)$ is written in the way shown in (1), we call it ordered according to descending powers of x . The summands may be arranged in any other order; for example, the function may also be ordered according to ascending powers of x :

$$f(x) = a_n + a_{n-1}x + \cdots + a_1x^{n-1} + a_0x^n.$$

By addition, subtraction, and multiplication of whole rational functions there again arise whole rational functions. The rules of literal calculation directly give the laws of formation of these new functions.

In addition, the coefficient of any power x^ν in the sum is equal to the sum of the coefficients of x^ν in the individual summands. The degree of the function that arises is equal to the highest of the degrees of the summands, and can decrease only in the special case in which the highest degree occurs in several summands and the sum of the coefficients of the highest powers is zero.

In the multiplication of two or more whole rational functions there arises a function whose degree is equal to the sum of the degrees of the factors.

To survey the law of formation of the coefficients for the product, put

$$\begin{aligned} A(x) &= a_0x^m + a_1x^{m-1} + a_2x^{m-2} + \cdots, \\ B(x) &= b_0x^n + b_1x^{n-1} + b_2x^{n-2} + \cdots, \end{aligned} \quad (2)$$

$$A(x)B(x) = C(x) = c_0x^{m+n} + c_1x^{m+n-1} + c_2x^{m+n-2} + \cdots \quad (3)$$

and obtain

$$\begin{aligned} c_0 &= a_0b_0, \\ c_1 &= a_0b_1 + a_1b_0, \\ c_2 &= a_0b_2 + a_1b_1 + a_2b_0, \\ &\cdots, \end{aligned}$$

or, in general notation, if ν denotes one of the numbers $0, 1, 2, \dots, m+n$,

$$c_\nu = a_0b_\nu + a_1b_{\nu-1} + a_2b_{\nu-2} + \cdots + a_{\nu-1}b_1 + a_\nu b_0, \quad (4)$$

where all a whose index is greater than m , and all b whose index is greater than n , are to be set equal to zero. For c_ν is the coefficient of $x^{m+n-\nu}$, and hence $c_\nu x^{m+n-\nu}$ arises by multiplying all terms of the form $a_\mu x^{m-\mu}$ by all terms of the form $b_{\nu-\mu} x^{n-\nu+\mu}$ and then adding. If a_0 and $b_0 = 1$, then also $c_0 = 1$.

If in all factors of such a product the highest power of x has coefficient 1, then in the product too the coefficient of the highest power is 1.

From these rules for calculating with whole functions it follows that the rules of calculation expressed in formulae such as $ab = ba$, $(ab)c = a(bc)$, $(a + b)c = ac + bc$ and the like also hold when a, b, c are whole functions of x , and in the sense that whole functions are to be regarded as equal only when equal powers have equal coefficients.

By whole functions of several variables we understand sums of products of whole, positive powers of the variables $x, y, z \dots$ with arbitrary coefficients that count as constant. One can think of these functions as arising from functions of one variable x when the coefficients $a_0, a_1 \dots a_n$ occurring in them are themselves regarded as whole rational functions of other variables $y, z \dots$. Thus functions of m variables arise from functions of $m - 1$ variables. All terms of such a function are of the form $x^r y^s z^t \dots$, multiplied by a coefficient; and if the function is properly collected, then each combination of the exponents $r, s, t \dots$ occurs only once. Two whole functions so ordered are regarded as equal only when they contain the same products $x^r y^s z^t \dots$ with the same coefficients.

§ 2. A theorem of Gauss.

We shall at once make an application of the multiplication rule for two whole rational functions, in order to prove a theorem of Gauss which will be useful to us later, but which here is to serve as an introduction to the mode of calculation and as an example.⁷

We consider here the case in which the coefficients in the functions $A(x), B(x)$ are whole numbers, so that, by (4), the coefficients of $C(x) = A(x)B(x)$ are also whole numbers.

If all integral coefficients $a_0, a_1, \dots, a_m$ of a function $A(x)$ have no common divisor, then the function $A(x)$ is called an original or primitive function, and the theorem we wish to prove is:

If $A(x)$ and $B(x)$ are primitive functions, then their product $C(x)$ is also a primitive function.

The proof follows almost immediately from looking at formula (4).

For if all coefficients $c_0, c_1, c_2 \dots c_{m+n}$, as given in (4), had a common divisor greater than 1, then there would also have to be at least one prime number which divides all these coefficients. Let p be such a prime number. By the assumption that $A(x), B(x)$ are primitive, this prime can divide neither all $a_0, a_1 \dots a_m$ nor all $b_0, b_1 \dots b_n$.

Suppose now that p divides

$$a_0, a_1 \dots a_{r-1}, \quad \text{but not } a_r,$$

and divides

$$b_0, b_1 \dots b_{s-1}, \quad \text{but not } b_s.$$

Then, by assumption, $r \leq m$, and r may also be zero if p already does not divide a_0 . Likewise $s \leq n$.

If now, by (4), we form c_{r+s} and arrange it in the following way,

$$\begin{aligned} c_{r+s} = & a_r b_s + a_{r-1} b_{s+1} + a_{r-2} b_{s+2} + \dots \\ & + a_{r+1} b_{s-1} + a_{r+2} b_{s-2} + \dots, \end{aligned} \tag{1}$$

one sees at once that c_{r+s} cannot be divisible by p , as had been assumed; for the first term $a_r b_s$ is not divisible by p , while all other terms, since they are multiplied by one of the coefficients $a_0, a_1 \dots a_{r-1}, b_0, b_1 \dots b_{s-1}$, are divisible by p . Thus the assumption that $C(x)$ is not primitive while $A(x)$ and $B(x)$ are primitive leads to a contradiction.

This theorem can be transferred to functions of several variables. We call a whole rational function of m variables with integral coefficients original or primitive if the coefficients have no common divisor, and we state the theorem that the product of two primitive functions is again a primitive function.

⁷Gauss, *Disquisitiones arithmeticae*, Art. 42.

To see its truth, we need only take, in the argument carried out above, the coefficients $a_0, a_1, a_2 \dots, b_0, b_1, b_2 \dots$ not as whole numbers, but as whole rational functions of $m - 1$ variables $y, z \dots$, and call such a function divisible by a prime number p when all its coefficients are divisible by p . If we then suppose that the theorem to be proved has already been proved for functions of $m - 1$ variables, formula (1) gives its correctness for functions of m variables, hence its general validity by complete induction, or by the conclusion from $m - 1$ to m .

An imprimitive function is one whose integral coefficients all have a common divisor. We call this divisor the divisor of the function. Then we can also express the theorem just proved as follows:

The divisor of a product of two whole functions is equal to the product of the divisors of the two functions.

For if PA and QB are two whole functions with divisors P and Q , then A and B are primitive functions. Hence $AB = C$ is also a primitive function, and PQ is the divisor of the function $PA \cdot QB = PQC$.

Insofar as it refers to functions of one variable, we can give the theorem the following form, without changing its content essentially; in this form it is especially useful. Let

$$\varphi(x) = x^m + \alpha_1 x^{m-1} + \alpha_2 x^{m-2} + \dots + \alpha_m,$$

$$\psi(x) = x^n + \beta_1 x^{n-1} + \beta_2 x^{n-2} + \dots + \beta_n$$

be two whole rational functions in which the highest powers of x have coefficient 1, while the remaining coefficients are rational numbers. Then, in the product

$$\varphi(x)\psi(x) = x^{m+n} + \gamma_1 x^{m+n-1} + \gamma_2 x^{m+n-2} + \dots + \gamma_{m+n},$$

the coefficients γ cannot all be whole numbers if the coefficients α, β in $\varphi(x)$ and $\psi(x)$ are not all whole numbers.

For let the smallest common denominator of the coefficients α of φ be denoted by a_0 , and that of the coefficients β of ψ by b_0 . Then $a_0\varphi(x) = A(x)$ and $b_0\psi(x) = B(x)$ are primitive functions of x . If the coefficients γ were whole numbers, the product $a_0b_0\varphi(x)\psi(x)$ would have divisor a_0b_0 and, if a_0 and b_0 were not both equal to 1, would therefore not be primitive; but this would contradict the theorem proved above.

§ 3. Division.

Let, as before,

$$\begin{aligned} A &= A(x) = a_0 x^m + a_1 x^{m-1} + \dots, \\ B &= B(x) = b_0 x^n + b_1 x^{n-1} + \dots \end{aligned} \tag{1}$$

be two whole rational functions of x ; but now suppose that $m \geq n$ and that a_0 and b_0 are different from zero. Then the difference

$$A - \frac{a_0}{b_0} x^{m-n} B \tag{2}$$

is also a whole rational function of x , but its degree is smaller than m , since the highest power in the two terms of the difference has the same coefficient and therefore cancels. We put this difference equal to

$$A' = A'(x) = a'_0 x^{m'} + a'_1 x^{m'-1} + \dots, \quad m' < m. \tag{3}$$

If now m' is still $\geq n$, then in (2) we can put A' in place of A and repeat the same reasoning.

This gives a chain of equations:

$$\begin{aligned} A - \frac{a_0}{b_0}x^{m-n}B &= A', \\ A' - \frac{a'_0}{b_0}x^{m'-n}B &= A'', \\ A'' - \frac{a''_0}{b_0}x^{m''-n}B &= A''', \\ &\dots, \end{aligned} \tag{4}$$

and this chain can be continued until the degree of the function that has arisen is smaller than n . Since in the sequence of functions $A, A', A'' \dots$ the degree of each following function is lowered by at least one unit, the chain of equations (4) consists of at most $m - n + 1$ members; but it may also contain fewer members when several powers cancel simultaneously. If we add all equations (4), denote the last of the functions $A, A' \dots$ by C , and, for abbreviation, set

$$Q = \frac{a_0}{b_0}x^{m-n} + \frac{a'_0}{b_0}x^{m'-n} + \dots, \tag{5}$$

so that Q also is a whole rational function of x , then there follows

$$A = QB + C. \tag{6}$$

The operation just described, by which the functions Q, C are found from A, B , is called division. A is the dividend, B the divisor, C the remainder, and Q the quotient. The degree of the remainder is always lower than the degree of the divisor.

The coefficients of the functions Q and C are composed from the coefficients a and b by addition, subtraction, multiplication, and division. In the denominator, however, only powers of b_0 occur; and hence if $b_0 = 1$, the coefficients of Q and C are whole functions of the a and b . The highest power of b_0 which can occur in the denominator is the $(m - n + 1)$ st, since in the chain (4) the factor b_0 is added once in the denominator in each following equation. In special cases, however, the highest power of b_0 in all denominators may be a lower one.

Take, for example, for A a function of the third degree,

$$f(x) = a_0x^3 + a_1x^2 + a_2x + a_3, \tag{7}$$

and for B the so-called first derivative of $f(x)$, which is of the second degree,

$$f'(x) = 3a_0x^2 + 2a_1x + a_2. \tag{8}$$

Then one obtains

$$Q = \frac{1}{3}x + \frac{a_1}{9a_0}, \tag{9}$$

$$C = \frac{6a_0a_2 - 2a_1^2}{9a_0}x + \frac{9a_0a_3 - a_1a_2}{9a_0}. \tag{10}$$

How the calculation prescribed in the equations (4) is expediently arranged may here be assumed as known from the elements. We draw attention to the analogy between this calculation and division in the decimal number system. A function $f(x)$ represents a number written decimally if the coefficients $a_0, a_1 \dots$ are whole numbers between zero, inclusive, and 10, exclusive, and $x = 10$ is set. If one also permits numbers greater than 10 among the coefficients, then one can represent a number by $f(x)$ in various ways. This is used in the division procedure in order to avoid fractional and negative coefficients in the results in the simplest way.

§ 4. Division by a linear function.

We shall apply the general rule for division to one special case. Let the dividend be

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n, \quad (1)$$

arbitrary, while the divisor is of the first degree, or, as one also says, linear. We shall also suppose the coefficient of the first power of x in the divisor to be 1, and therefore take the divisor in the simple form $(x - \alpha)$, where α remains arbitrary. In this case the quotient Q is of degree $n - 1$, and the remainder C of degree 0, that is, independent of x . Thus set

$$f(x) = (x - \alpha)Q + C. \quad (2)$$

Then C no longer contains x , and if we set

$$Q = q_0x^{n-1} + q_1x^{n-2} + \cdots + q_{n-2}x + q_{n-1}, \quad (3)$$

there follows from (2)

$$\begin{aligned} f(x) &= q_0x^n + q_1x^{n-1} + \cdots + q_{n-2}x^2 + q_{n-1}x + C \\ &\quad - \alpha q_0x^{n-1} - \cdots - \alpha q_{n-3}x^2 - \alpha q_{n-2}x - \alpha q_{n-1}, \end{aligned} \quad (4)$$

and by comparison with (1):

$$\begin{aligned} q_0 &= a_0, \\ q_1 - \alpha q_0 &= a_1, \\ q_2 - \alpha q_1 &= a_2, \\ &\cdots, \\ q_{n-1} - \alpha q_{n-2} &= a_{n-1}, \\ C - \alpha q_{n-1} &= a_n. \end{aligned} \quad (5)$$

From this one obtains

$$\begin{aligned} q_0 &= a_0, \\ q_1 &= a_0\alpha + a_1, \\ q_2 &= a_0\alpha^2 + a_1\alpha + a_2, \\ &\cdots, \\ q_{n-1} &= a_0\alpha^{n-1} + a_1\alpha^{n-2} + \cdots + a_{n-1}, \\ C &= a_0\alpha^n + a_1\alpha^{n-1} + \cdots + a_{n-1}\alpha + a_n = f(\alpha). \end{aligned} \quad (6)$$

C arises from $f(x)$ by setting $x = \alpha$, and therefore can also be denoted by $f(\alpha)$.

Accordingly we also have the formula

$$\frac{f(x) - f(\alpha)}{x - \alpha} = Q(x), \quad (7)$$

where $Q(x)$ is a whole function of degree $n - 1$.

If in the expressions (6) we put the sign x in place of the indeterminate quantity α , there arises from this a sequence of whole rational functions of x , which, if for simplicity we put $a_0 = 1$, we write as follows:

$$\begin{aligned} f_0 &= 1, \\ f_1 &= x + a_1, \\ f_2 &= x^2 + a_1x + a_2, \\ &\cdots, \\ f_{n-1} &= x^{n-1} + a_1x^{n-2} + a_2x^{n-3} + \cdots + a_{n-1}. \end{aligned} \quad (8)$$

These functions $f_0, f_1 \dots f_{n-1}$ will later render us good service. For the present we add the following remarks.

According to (8), one can express the powers $1, x, x^2 \dots x^{n-1}$ of x linearly in terms of the functions $f_0, f_1 \dots f_{n-1}$, and indeed in such a way that only integral rational combinations of the a occur in the coefficients, for instance

$$\begin{aligned} 1 &= f_0, \\ x &= f_1 - a_1 f_0, \\ x^2 &= f_2 - a_1 f_1 + (a_1^2 - a_2) f_0, \\ &\dots \end{aligned}$$

From this it follows that every whole rational function of x whose degree is not greater than $n - 1$ can likewise be expressed linearly by $f_0, f_1, \dots, f_{n-1}$ in the form

$$y_0 f_0 + y_1 f_1 + \dots + y_{n-1} f_{n-1}, \quad (9)$$

where the coefficients $y_0, y_1 \dots y_{n-1}$ are independent of x .

Thus, if $F(x)$ is any whole rational function of x , then by § 3, regarding $f(x)$ as divisor, one can set

$$F(x) = Qf(x) + y_0 f_0 + y_1 f_1 + \dots + y_{n-1} f_{n-1}, \quad (10)$$

where Q is also a whole rational function of x .

For the recurrent calculation of the functions $f_r(x)$, relation (8) gives

$$f_r(x) - x f_{r-1}(x) = a_r. \quad (11)$$

§ 5. Rational fractional functions; divisibility.

If $F(x)$ and $f(x)$ are two integral rational functions of x , then the fraction

$$\frac{F(x)}{f(x)} \quad (1)$$

is called a *fractional rational*, or also briefly a *fractional* or *rational function* of x .

If the degree of the numerator is lower than the degree of the denominator, then the function is called a *proper fraction*; in the opposite case it is called an *improper fraction*.

According to § 3 the integral rational functions Q and $\varphi(x)$ can be so determined that

$$F(x) = Qf(x) + \varphi(x) \quad (2)$$

and the degree of $\varphi(x)$ is smaller than the degree of $f(x)$; consequently

$$\frac{F(x)}{f(x)} = Q + \frac{\varphi(x)}{f(x)}, \quad (3)$$

and hence the theorem:

Every fractional function can be decomposed into the sum of an integral function and a proper fractional function.

If n is the degree of $f(x)$, then the degree of $\varphi(x)$ is at most $n - 1$; it may also be lower. In particular, the case may occur in which $\varphi(x)$ vanishes identically.

In this case the function $F(x)$ is said to be divisible by $f(x)$. The function

$$\frac{F(x)}{f(x)}$$

is in this case only apparently fractional; in reality it is equal to the integral function Q .

The formula

$$\frac{x^m - 1}{x - 1} = x^{m-1} + x^{m-2} + x^{m-3} + \cdots + 1$$

gives a simple example of this.

For divisibility of functions the same laws hold as for divisibility of numbers, in particular the following:

1. If the function $F(x)$ is divisible by the function $f(x)$, and $f(x)$ is divisible by a third function $\varphi(x)$, then $F(x)$ is also divisible by $\varphi(x)$.

For if

$$F = Qf, \quad f = q\varphi,$$

where Q and q are integral rational functions, then

$$F = Qq\varphi,$$

and since Qq is an integral rational function, F is divisible by φ .

2. If $F(x)$ is divisible by $f(x)$ and Q is any integral rational function, then $QF(x)$ is also divisible by $f(x)$.

3. If $F(x)$ and $f(x)$ are divisible by $\varphi(x)$, then $F(x) \pm f(x)$ is also divisible by $\varphi(x)$, or more generally:

4. If $F_1, F_2, \dots$ are divisible by $f(x)$ and $Q_1, Q_2, \dots$ are arbitrary integral rational functions, then

$$Q_1F_1 + Q_2F_2 + \cdots$$

is also divisible by $f(x)$.

The last theorem includes the two preceding ones and is proved simply as follows.

If $F_1, F_2 \dots$ are divisible by f , then the integral rational functions $\Phi_1, \Phi_2 \dots$ can be determined so that

$$F_1 = \Phi_1 f, \quad F_2 = \Phi_2 f, \dots$$

and consequently

$$Q_1 F_1 + Q_2 F_2 + \dots = (Q_1 \Phi_1 + Q_2 \Phi_2 + \dots) f.$$

Since $Q_1 \Phi_1 + Q_2 \Phi_2 + \dots$ is an integral rational function, the theorem is proved.

5. Every function is divisible by itself.

With respect to divisibility or indivisibility of functions nothing is changed if the functions are multiplied by arbitrary factors independent of x .

A non-zero quantity independent of x may be regarded as a function of degree zero. If we call such a quantity a constant, we may say:

6. Every function is divisible by every constant.

If a function is divisible by another, then the degree of the quotient is equal to the difference between the degree of the dividend and the degree of the divisor. The former therefore cannot be smaller than the latter. If the degrees are equal, the quotient is a constant, and hence:

7. If, of two integral rational functions of the same degree, one is divisible by the other, then they differ from one another only by a constant factor, and the second is also divisible by the first.

8. By § 4, the necessary and sufficient condition that the function $f(x)$ be divisible by the linear function $x - \alpha$ is that $f(\alpha) = 0$.

§ 6. Greatest common divisor.

It is a problem of fundamental importance to decide whether two integral rational functions have, besides the constants, any further common divisor. The solution is found by the algorithm of the greatest common divisor in exactly the same way as the corresponding question concerning divisibility of integers is answered. See the Introduction.

Let

$$f(x) = A, \quad \varphi(x) = A' \tag{1}$$

be two given functions; the degree of $\varphi(x)$ is supposed to be lower, or at least not higher, than that of $f(x)$.

We divide A by A' and denote the remainder, whose degree is lower than the degree of A' , by A'' , thus:

$$A = Q' A' + A'';$$

then we divide A' by A'' and denote the remainder by A''' , and continue in this way in forming the sequence of functions

$$A, A', A'', A''', \dots, \tag{2}$$

whose degrees $n, n', n'', n''' \dots$ always decrease and therefore, after a finite number of divisions, necessarily go down to zero. Let the last remainder of degree zero, and therefore a constant, be $A^{(\nu)}$. Then we have the chain of equations

$$\begin{aligned} A &= Q' A' + A'', \\ A' &= Q'' A'' + A''', \\ &\dots \\ A^{(\nu-3)} &= Q^{(\nu-2)} A^{(\nu-2)} + A^{(\nu-1)}, \\ A^{(\nu-2)} &= Q^{(\nu-1)} A^{(\nu-1)} + A^{(\nu)}, \end{aligned} \tag{3}$$

where $A^{(\nu)}$ is a constant.

If now A, A' have any common divisor, then by the theorems of the preceding paragraph, as is seen from the equations (3), it is also a divisor of A'', A''', A'''' and so on down to $A^{(\nu)}$. If $A^{(\nu)}$ is a non-zero constant, then no divisor of $f(x)$ and $\varphi(x)$ depending on x can exist. Such functions are called *prime to one another*, or *relatively prime*. One also says, taking account only of divisors depending on x , that the functions have no common divisor.

The condition that the two functions have a common divisor is therefore

$$A^{(\nu)} = 0. \quad (4)$$

In this case $A^{(\nu-1)}$ is a divisor of $A^{(\nu-2)}$, as the last equation (3) shows, and by the preceding equation it is a divisor of $A^{(\nu-3)}$, and so on, hence also a divisor of A and A' , that is, of f and φ . Conversely, every common divisor of A, A' is also a divisor of $A^{(\nu-1)}$; accordingly $A^{(\nu-1)}$ is called the *greatest common divisor* of f and φ . The algorithm (3) shows that $A^{(\nu)}$ or $A^{(\nu-1)}$ can be derived from the coefficients of f and φ by the rational operations of arithmetic, and in such a manner that division is always only by the coefficients of the highest powers of x in the functions $A, A', A'', \dots$ which are not zero.

We shall apply these considerations to two examples.

First let

$$\begin{aligned} A &= a_0x^2 + a_1x + a_2, \\ B &= b_0x^2 + b_1x + b_2 \end{aligned} \quad (5)$$

be two functions of the second degree, so that a_0, b_0 are non-zero.

The first step is to form the equation

$$A = \frac{a_0}{b_0}B + C, \quad (6)$$

where

$$C = c_0x + c_1 \quad (7)$$

and

$$c_0 = \frac{a_1b_0 - b_1a_0}{b_0}, \quad c_1 = \frac{a_2b_0 - b_2a_0}{b_0}. \quad (8)$$

If $c_0 = 0$, so that C is constant, then A, B have a common factor only if also $c_1 = 0$; and then A is divisible by B , that is, A and B differ only by a constant factor. If however c_0 is different from zero, we continue the calculation by putting

$$B = QC + D,$$

where, according to § 4, if there $\alpha = -c_1 : c_0$ is put,

$$D = \frac{b_0c_1^2 - b_1c_0c_1 + b_2c_0^2}{c_0^2}. \quad (9)$$

If D is non-zero, then A, B have no common divisor. If however $D = 0$, then C is the greatest common factor of A and B .

Substituting the values c_0, c_1 from (8) into (9), and suppressing the denominator $b_0c_0^2$, one obtains the condition for the existence of a common divisor C of A and B in the form

$$\begin{aligned} &a_0^2b_2^2 + a_2^2b_0^2 - 2a_0a_2b_0b_2 - a_1a_2b_0b_1 - a_0a_1b_1b_2 \\ &+ a_0a_2b_1^2 + a_1^2b_0b_2 = 0 \end{aligned} \quad (10)$$

or

$$(a_0b_2 - b_0a_2)^2 + (a_0b_1 - a_1b_0)(a_2b_1 - a_1b_2) = 0. \quad (11)$$

This condition is also fulfilled when c_0 and c_1 are both zero, and is therefore the necessary and sufficient condition that A, B have a common divisor.

The left hand side of (10) or (11) is called the *resultant* of the functions A and B .

As a second example we take the one already chosen in § 3. Put

$$\begin{aligned} f(x) &= a_0x^3 + a_1x^2 + a_2x + a_3 = A, \\ f'(x) &= 3a_0x^2 + 2a_1x + a_2 = B, \end{aligned} \quad (12)$$

and suppose a_0 different from zero. Then by § 3 we have

$$A = QB + C, \quad (13)$$

$$C = c_0x + c_1, \quad (14)$$

where

$$c_0 = \frac{6a_0a_2 - 2a_1^2}{9a_0}, \quad c_1 = \frac{9a_0a_3 - a_1a_2}{9a_0}. \quad (15)$$

If c_0 is zero, the algorithm is already closed. If c_1 is non-zero, then A and B have no common divisor; but if c_1 is also zero, then B itself is the greatest common divisor of A and B , that is, A is divisible by B . The conditions for this are therefore

$$3a_0a_2 - a_1^2 = 0, \quad 9a_0a_3 - a_1a_2 = 0. \quad (16)$$

If c_0 is not zero, we go one step further and put

$$B = PC + D, \quad (17)$$

where D becomes constant and has the expression

$$D = \frac{a_2c_0^2 - 2a_1c_0c_1 + 3a_0c_1^2}{c_0^2}. \quad (18)$$

If this expression is non-zero, then A and B are prime to one another; if it is zero, then A and B have the greatest common divisor C . If for c_0, c_1 one substitutes the values (15), omits the denominator, still removes the non-vanishing factor $9a_0$, and changes the sign, this condition, after a simple calculation, assumes the form

$$a_1^2a_2^2 + 18a_0a_1a_2a_3 - 4a_0a_2^3 - 4a_1^3a_3 - 27a_0^2a_3^2 = 0. \quad (19)$$

It is, as one readily sees either by calculation or from (18), also fulfilled when the conditions (16) hold, and is therefore the necessary and sufficient condition that $f(x)$ and $f'(x)$ have a common divisor. The left hand side of (19), which is an integral rational and homogeneous function of the coefficients of $f(x)$, is called the *discriminant* of the function $f(x)$.

We derive one more theorem from the algorithm (3), which is often used.

From the first of these equations it follows that

$$A'' = A - Q'A', \quad (20)$$

and if this value of A'' is substituted in the following equation, one obtains

$$A''' = (1 + Q'Q'')A' - Q''A,$$

thus, denoting by p, p' integral rational functions,

$$A''' = pA + p'A'. \quad (21)$$

Substituting the expressions (20), (21) in the third equation (3), one obtains for A'''' again an expression of the form (21), and so one may continue, obtaining finally

$$A^{(\nu)} = PA + P'A', \quad (22)$$

where P, P' are integral rational functions whose coefficients are composed from the coefficients of A and A' by rational operations.

In the formula (22), $A^{(\nu)}$ is a constant. This theorem is especially important in the case where $A^{(\nu)}$ is non-zero, so that A, A' are relatively prime. If in this case we put

$$P = A^{(\nu)}F(x), \quad P' = A^{(\nu)}\Phi(x),$$

then, after suppressing the factor $A^{(\nu)}$, the theorem can be expressed as follows:

I. If $f(x)$ and $\psi(x)$ are two integral functions without a common divisor, then two other integral functions $F(x)$ and $\Phi(x)$ can be determined which satisfy identically the equation

$$F(x)f(x) + \Phi(x)\psi(x) = 1. \quad (23)$$

The theorem can still be generalized. Multiplying (23) by an arbitrary integral function $\chi(x)$ gives

$$F(x)\chi(x)f(x) + \Phi(x)\chi(x)\psi(x) = \chi(x), \quad (24)$$

and by § 3 we can put

$$\Phi(x)\chi(x) = Q(x)f(x) + \varphi(x) \quad (25)$$

in such a way that $Q(x), \varphi(x)$ are integral rational functions of x and the degree of $\varphi(x)$ is lower than that of $f(x)$. Substituting this in (24), and putting in place of $F(x)\chi(x) + Q(x)\psi(x)$ again $F(x)$, we obtain

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x).$$

We can therefore generalize the preceding theorem as follows:

II. If $f(x), \psi(x), \chi(x)$ are given integral rational functions, and $f(x)$ and $\psi(x)$ are without a common divisor, then the integral rational functions $F(x), \varphi(x)$ can be so determined - with $\varphi(x)$ of lower degree than $f(x)$ - that the equation

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x) \quad (26)$$

is identically satisfied.

§ 7. Products of linear factors.

According to § 1, by multiplication of integral functions we again obtain such functions of higher degree, and the degree of the product is determined by the sum of the degrees of the individual factors.

If therefore we multiply n linear factors by one another, there arises a function of the n th degree, whose manner of formation we must examine somewhat more closely.

We shall take the linear factors in the simple form $x - \alpha_1, x - \alpha_2, \dots, x - \alpha_n$, and put, since the coefficient of the highest power is, by § 1, equal to 1,

$$\begin{aligned} f(x) &= (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n) \\ &= x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n. \end{aligned} \quad (1)$$

A slight consideration shows the following law of formation of the coefficients $a_1, a_2, \dots, a_n$:

$-a_1$ is equal to the sum of the α 's; a_2 is equal to the sum of the products of every two of the α 's; $-a_3$ is the sum of the products of every three of the α 's; in general $(-1)^\nu a_\nu$ is the sum of the

products of every ν of the quantities α , or, in formulas,

$$\begin{aligned}
 -a_1 &= \sum \alpha_1, \\
 +a_2 &= \sum \alpha_1 \alpha_2, \\
 &\dots \\
 (-1)^\nu a_\nu &= \sum \alpha_1 \alpha_2 \cdots \alpha_\nu, \\
 &\dots \\
 (-1)^n a_n &= \alpha_1 \alpha_2 \cdots \alpha_n.
 \end{aligned} \tag{2}$$

But to see still more clearly the correctness of this law of formation, one uses complete induction.

One first verifies the correctness in the first cases by actually carrying out the multiplication:

$$\begin{aligned}
 (x - \alpha_1)(x - \alpha_2) &= x^2 - (\alpha_1 + \alpha_2)x + \alpha_1 \alpha_2, \\
 (x - \alpha_1)(x - \alpha_2)(x - \alpha_3) &= x^3 - (\alpha_1 + \alpha_2 + \alpha_3)x^2 \\
 &\quad + (\alpha_1 \alpha_2 + \alpha_1 \alpha_3 + \alpha_2 \alpha_3)x - \alpha_1 \alpha_2 \alpha_3.
 \end{aligned}$$

If we assume the law of formation proved for $n - 1$ factors, and put

$$(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_{n-1}) = x^{n-1} + a'_1 x^{n-2} + \cdots + a'_{n-1},$$

then multiplication by $x - \alpha_n$ gives

$$\begin{aligned}
 a_1 &= a'_1 - \alpha_n, \\
 a_2 &= a'_2 - \alpha_n a'_1, \\
 a_3 &= a'_3 - \alpha_n a'_2, \\
 &\dots \\
 a_\nu &= a'_\nu - \alpha_n a'_{\nu-1}, \\
 &\dots \\
 a_n &= -\alpha_n a'_{n-1},
 \end{aligned} \tag{3}$$

and from this the correctness of the formulas (2) is immediately evident.

It is important to determine the number of terms which occur in each of the sums (2). We denote the number of terms occurring in the sum $(-1)^\nu a_\nu$ by $B_\nu^{(n)}$; it is the number of combinations without repetition of n elements to the ν th class, that is, of every ν distinct ones among the n elements. To determine it, suppose first that the $B_{\nu-1}^{(n)}$ combinations to the $(\nu - 1)$ st class have been formed. From each of these combinations, by adding one at a time of the missing $n - \nu + 1$ elements, one can derive $n - \nu + 1$ combinations to the ν th class. In this way however every combination is formed ν times, namely by addition of each of its elements; hence one obtains the relation

$$B_\nu^{(n)} = B_{\nu-1}^{(n)} \frac{n - \nu + 1}{\nu}, \tag{4}$$

whereas $B_1^{(n)}$ plainly has the value n . Therefore, if one multiplies the equations following from (4),

$$\begin{aligned}
 B_1^{(n)} &= n, \\
 B_2^{(n)} &= B_1^{(n)} \frac{n-1}{2}, \\
 &\dots \\
 B_\nu^{(n)} &= B_{\nu-1}^{(n)} \frac{n - \nu + 1}{\nu},
 \end{aligned} \tag{5}$$

it follows that

$$B_\nu^{(n)} = \frac{n(n-1)(n-2)\cdots(n-\nu+1)}{1\cdot 2\cdot 3\cdots \nu}. \quad (6)$$

In what follows we shall often, when m is an arbitrary positive integer, use the symbol

$$\Pi(m) = 1\cdot 2\cdot 3\cdots m, \quad \Pi(0) = 1, \quad (7)$$

so that

$$\Pi(m) = m\Pi(m-1). \quad (8)$$

With the help of this symbol the expression for $B_\nu^{(n)}$ can be displayed more conveniently as

$$B_\nu^{(n)} = B_{n-\nu}^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)}, \quad (9)$$

which, if $B_0^{(n)} = 1$ is put, is also valid for $\nu = 0$ and $\nu = n$, and makes evident the invariance of $B_\nu^{(n)}$ under the interchange of ν with $n - \nu$.

The derivation of the formula (4), which we have just given according to the rules of the theory of combinations, is indeed perfectly correct and clear, but for its exact foundation it requires some reflection which cannot easily be compressed into a few words. We shall therefore subsequently prove its correctness once more by the method of complete induction.

The formulas (3), namely, give the following recurrence formulas:

$$\begin{aligned} B_1^{(n)} &= B_1^{(n-1)} + 1, \\ B_2^{(n)} &= B_2^{(n-1)} + B_1^{(n-1)}, \\ &\dots \\ B_\nu^{(n)} &= B_\nu^{(n-1)} + B_{\nu-1}^{(n-1)}, \\ &\dots \\ B_n^{(n)} &= B_{n-1}^{(n-1)}. \end{aligned} \quad (10)$$

Now the formula (6) can be confirmed very easily by counting for the first values of n . If one assumes it correct for $n - 1$, then (10) gives

$$B_\nu^{(n)} = \frac{\Pi(n-1)}{\Pi(\nu)\Pi(n-\nu-1)} + \frac{\Pi(n-1)}{\Pi(\nu-1)\Pi(n-\nu)},$$

whence, by (8),

$$B_\nu^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)},$$

and by this the general validity of formula (9) is proved.

From the meaning of $B_\nu^{(n)}$ it follows, and it is also proved from the formulas (10) by complete induction, that the $B_\nu^{(n)}$ are positive integers.

§ 8. The binomial theorem.

If in formula (1) of the preceding paragraph we set the previously arbitrary quantities $\alpha_1, \alpha_2 \dots \alpha_n$ equal to one another, we obtain the binomial theorem, which consists in nothing other than arranging the n th power of a binomial $x + y$ according to powers of x . Namely, if we put

$$\alpha_1 = \alpha_2 = \cdots = \alpha_n = -y,$$

then formula (2) gives

$$a_\nu = (-1)^\nu \sum \alpha_1 \alpha_2 \cdots \alpha_\nu = y^\nu B_\nu^{(n)},$$

where, according to the definition of the preceding paragraph, $B_\nu^{(n)}$ means the number of terms of the sum, which all become equal to one another and equal to $(-1)^\nu y^\nu$.

Accordingly one obtains

$$(x + y)^n = x^n + B_1^{(n)} x^{n-1} y + B_2^{(n)} x^{n-2} y^2 + \cdots + B_n^{(n)} y^n, \quad (1)$$

or, written out,

$$\begin{aligned} (x + y)^n &= x^n + n x^{n-1} y + \frac{n(n-1)}{1 \cdot 2} x^{n-2} y^2 + \cdots + y^n \\ &= \Pi(n) \sum \frac{x^{n-\nu} y^\nu}{\Pi(n-\nu) \Pi(\nu)} = \Pi(n) \sum \frac{x^\alpha y^\beta}{\Pi(\alpha) \Pi(\beta)}, \end{aligned}$$

the last sum being extended over all non-negative integral values α, β which satisfy the condition $\alpha + \beta = n$.

Accordingly the coefficients $B_\nu^{(n)}$ are called the binomial coefficients.

For reference we put here a small table of the first values of the binomial coefficients:

$n = 1$	1	1							
$n = 2$	1	2	1						
$n = 3$	1	3	3	1					
$n = 4$	1	4	6	4	1				
$n = 5$	1	5	10	10	5	1			
$n = 6$	1	6	15	20	15	6	1		
$n = 7$	1	7	21	35	35	21	7	1	

Among the various properties of the binomial coefficients we shall derive two, of which we shall afterwards make an interesting application.

From (1), replacing x, y by $1, x$ and taking $n = 0, 1, 2, 3 \dots$, the formulas

$$\begin{aligned} 1 &= B_0^{(0)}, \\ 1 + x &= B_0^{(1)} + B_1^{(1)} x, \\ (1 + x)^2 &= B_0^{(2)} + B_1^{(2)} x + B_2^{(2)} x^2, \\ &\dots \\ (1 + x)^n &= B_0^{(n)} + B_1^{(n)} x + B_2^{(n)} x^2 + \cdots + B_n^{(n)} x^n \end{aligned} \quad (2)$$

are obtained.

We now use the well-known summation formula for the geometric series:

$$1 + (1 + x) + (1 + x)^2 + \cdots + (1 + x)^n = \frac{(1 + x)^{n+1} - 1}{x},$$

where, if $(1 + x)^{n+1}$ is again expanded by the binomial formula, replacing n by $n + 1$ in the last formula (2) and setting $B_0^{(n+1)} = 1$,

$$\frac{(1 + x)^{n+1} - 1}{x} = B_1^{(n+1)} + B_2^{(n+1)} x + \cdots + B_{n+1}^{(n+1)} x^n. \quad (3)$$

Comparing this with the sum of the right hand sides of (2), and, according to the rules of § 1, setting equal the coefficients of corresponding powers of x , one obtains

$$\begin{aligned} B_0^{(0)} + B_0^{(1)} + B_0^{(2)} + \cdots + B_0^{(n)} &= B_1^{(n+1)}, \\ B_1^{(1)} + B_1^{(2)} + \cdots + B_1^{(n)} &= B_2^{(n+1)}, \\ &\dots \\ B_n^{(n)} &= B_{n+1}^{(n+1)}, \end{aligned} \quad (4)$$

or in general

$$B_\nu^{(\nu)} + B_\nu^{(\nu+1)} + \cdots + B_\nu^{(n)} = B_{\nu+1}^{(n+1)}. \quad (5)$$

If however the equations (2) are multiplied in order by

$$B_0^{(n)}, \quad -B_1^{(n)}, \quad +B_2^{(n)}, \dots, \quad \pm B_n^{(n)},$$

where the upper sign is to be taken for even n and the lower for odd n , the sum of the left hand sides gives

$$B_0^{(n)} - B_1^{(n)}(1+x) + B_2^{(n)}(1+x)^2 - \cdots \pm B_n^{(n)}(1+x)^n = [1 - (1+x)]^n = (-x)^n,$$

and equating the coefficients of equal powers on the right and left hand sides yields the system of formulas

$$\begin{aligned} 0 &= B_0^{(0)} B_0^{(n)} - B_0^{(1)} B_1^{(n)} + B_0^{(2)} B_2^{(n)} - \cdots \pm B_0^{(n)} B_n^{(n)}, \\ 0 &= -B_1^{(1)} B_1^{(n)} + B_1^{(2)} B_2^{(n)} - \cdots \pm B_1^{(n)} B_n^{(n)}, \\ &\dots \\ \pm 1 &= \pm B_n^{(n)} B_n^{(n)}, \end{aligned} \quad (6)$$

and this system of formulas can also be written in the reversed order, using the relation $B_\nu^{(n)} = B_{n-\nu}^{(n)}$, as

$$\begin{aligned} 1 &= B_0^{(n)} B_0^{(n)}, \\ 0 &= B_1^{(n)} B_0^{(n)} - B_0^{(n-1)} B_1^{(n)}, \\ 0 &= B_2^{(n)} B_0^{(n)} - B_1^{(n-1)} B_1^{(n)} + B_0^{(n-2)} B_2^{(n)}, \\ &\dots \\ 0 &= B_n^{(n)} B_0^{(n)} - B_{n-1}^{(n-1)} B_1^{(n)} + B_{n-2}^{(n-2)} B_2^{(n)} - \cdots \pm B_0^{(0)} B_n^{(n)}, \end{aligned} \quad (7)$$

or, in a summarizing notation,

$$\sum_{\nu=0}^{\mu} (-1)^\nu B_{\mu-\nu}^{(n-\nu)} B_\nu^{(n)} = \begin{cases} 1, & \mu = 0, \\ 0, & \mu = 1, 2, \dots, n. \end{cases} \quad (8)$$

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§ 9. Interpolation.

We shall make an application of the last obtained formulae to a problem from the theory of interpolation.

A whole rational function of degree n is to be determined which, for $n + 1$ given values of the variable x , has prescribed values.

Thus the problem is the determination of the $n + 1$ coefficients $a_0, a_1, \dots, a_n$ in

$$f(x) = a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n \quad (1)$$

from the $n + 1$ equations

$$f(\alpha_0) = A_0, \quad f(\alpha_1) = A_1, \quad \dots, \quad f(\alpha_n) = A_n, \quad (2)$$

when $\alpha_0, \alpha_1, \dots, \alpha_n$ are the given values of x , and $A_0, A_1, \dots, A_n$ are the corresponding values of $f(x)$.

Equations (2) form a system of equations of the first degree for the unknowns $a_0, a_1, \dots, a_n$, which in general can be solved by determinants, as we shall see in the next section. We shall later return to this problem, but treat it now under the special supposition that the given values of x are the first $n + 1$ whole numbers $0, 1, 2, \dots, n$.

The binomial coefficients $B_\nu^{(x)}$, as they were defined by § 7, (6), keep their good sense even when x is not a whole number, as was previously presupposed, but an arbitrary variable quantity. Then

$$B_\nu^{(x)} = \frac{x(x-1) \cdots (x-\nu+1)}{1 \cdot 2 \cdot 3 \cdots \nu} \quad (3)$$

is a whole rational function of degree ν in x .⁸ The coefficient of x^ν is different from zero, and therefore x^ν can also be expressed by $B_\nu^{(x)}$ and by lower powers of x . Thus every whole function of degree n in x can also be linearly expressed by $B_0^{(x)}, B_1^{(x)}, \dots, B_n^{(x)}$ in the form

$$f(x) = M_0B_0^{(x)} + M_1B_1^{(x)} + \dots + M_nB_n^{(x)}, \quad (4)$$

where $M_0, M_1, \dots, M_n$ are constants; and the function $f(x)$ is determined when these constants are determined.

Now, by our supposition, $f(0), f(1), f(2), \dots, f(n)$ are given. Since, by (3), $B_\nu^{(x)}$ always vanishes when x has one of the values $0, 1, 2, \dots, \nu - 1$, the following linear equations for the unknowns M are obtained from (4):

$$\begin{aligned} f(0) &= M_0B_0^{(0)}, \\ f(1) &= M_0B_0^{(1)} + M_1B_1^{(1)}, \\ f(2) &= M_0B_0^{(2)} + M_1B_1^{(2)} + M_2B_2^{(2)}, \\ &\dots \\ f(n) &= M_0B_0^{(n)} + M_1B_1^{(n)} + \dots + M_nB_n^{(n)}. \end{aligned} \quad (5)$$

These equations are now to be solved with respect to $M_0, M_1, \dots, M_n$, which is done very easily with the help of the final equations (8) of the preceding paragraph. The first equation (5) gives directly

$$M_0 = f(0). \quad (6)$$

⁸The meaning of these generalized binomial coefficients for the expansion of powers of a binomial does not belong here, but to analysis.

If the first equation (5) is multiplied by $B_0^{(1)}$, the second by $-B_1^{(1)}$, and they are added, one obtains, by the formula system mentioned (applied to $n = 1$),

$$-M_1 = B_0^{(1)}f(0) - B_1^{(1)}f(1),$$

and so in general, if the first ν equations (5) are multiplied in order by

$$B_0^{(\nu)}, \quad -B_1^{(\nu)}, \quad +B_2^{(\nu)}, \quad \dots, \quad \pm B_\nu^{(\nu)}$$

and then added,

$$\pm M_\nu = B_0^{(\nu)}f(0) - B_1^{(\nu)}f(1) + B_2^{(\nu)}f(2) - \dots \pm B_\nu^{(\nu)}f(\nu), \quad (7)$$

whereby, according to (4), the function $f(x)$ is determined and the problem is solved. It is clear that, so long as we make no special supposition about the values $f(0), f(1), \dots, f(n)$, every arbitrary whole rational function of x can be represented in this form.

§ 10. Solution of the interpolation problem by differences.

The definition of the whole rational function

$$B_\nu^{(x)} = \frac{x(x-1)\cdots(x-\nu+1)}{1 \cdot 2 \cdot 3 \cdots \nu} \quad (1)$$

gives the relation, already proved earlier for the special case of a positive integral x ,

$$B_\nu^{(x+1)} = B_\nu^{(x)} + B_{\nu-1}^{(x)}, \quad B_0^{(x+1)} = B_0^{(x)} = 1. \quad (2)$$

From this we obtain for the coefficients $M_0, M_1, \dots, M_n$ a mode of determination which is much more convenient for practical calculation than the application of the formulae of the preceding paragraph.

Namely, by the formulae (4) and (6) of § 9,

$$f(x) = f(0) + M_1 B_1^{(x)} + M_2 B_2^{(x)} + \dots + M_n B_n^{(x)}, \quad (3)$$

and if in this expression we replace x by $x+1$ and form the difference

$$\Delta_x = f(x+1) - f(x), \quad (4)$$

then, with regard to (2),

$$\Delta_x = M_1 + M_2 B_1^{(x)} + M_3 B_2^{(x)} + \dots + M_n B_{n-1}^{(x)}, \quad (5)$$

from which follows (since (5) is an equation of the same kind as (3), except that $n-1$ has taken the place of n)

$$M_1 = \Delta_0 = f(1) - f(0).$$

Put

$$\Delta'_x = \Delta_{x+1} - \Delta_x, \quad \Delta''_x = \Delta'_{x+1} - \Delta'_x, \quad \dots, \quad (6)$$

then, accordingly,

$$M_0 = f(0), \quad M_1 = \Delta_0, \quad M_2 = \Delta'_0, \quad \dots, \quad M_n = \Delta_0^{(n-1)},$$

and formula (3) gives

$$f(x) = f(0) + \Delta_0 B_1^{(x)} + \Delta'_0 B_2^{(x)} + \Delta''_0 B_3^{(x)} + \dots + \Delta_0^{(n-1)} B_n^{(x)}. \quad (7)$$

Likewise the functions $\Delta_x, \Delta'_x, \Delta''_x, \dots$ can be expressed:

$$\begin{aligned}\Delta_x &= \Delta_0 + \Delta'_0 B_1^{(x)} + \Delta''_0 B_2^{(x)} + \dots + \Delta_0^{(n-1)} B_{n-1}^{(x)}, \\ \Delta'_x &= \Delta'_0 + \Delta''_0 B_1^{(x)} + \dots + \Delta_0^{(n-1)} B_{n-2}^{(x)},\end{aligned}\tag{8}$$

and so on.

The $\Delta_x, \Delta'_x, \Delta''_x, \dots, \Delta_x^{(n-1)}$ are whole rational functions of degrees $n-1, n-2, \dots, 0$; the last of them is therefore constant. In order to represent them all, one needs only the values

$$f(0), \quad \Delta_0, \quad \Delta'_0, \quad \Delta''_0, \quad \dots, \quad \Delta_0^{(n-1)},$$

which one computes most easily by arranging a table. For the case $n=3$, for example, the table has the form

$$\begin{array}{cccc} f(0) & & & \\ f(1) & \Delta_0 & & \\ f(2) & \Delta_1 & \Delta'_0 & \\ f(3) & \Delta_2 & \Delta'_1 & \Delta''_0 \end{array}$$

where the differences are to be formed diagonally.

§ 11. Arithmetic series of higher order.

The preceding considerations can also be stated in another form, which we shall need later. If

$$u_0, u_1, u_2, \dots\tag{1}$$

is a sequence of quantities, we call

$$\Delta_0 = u_1 - u_0, \Delta_1 = u_2 - u_1, \Delta_2 = u_3 - u_2, \dots\tag{2}$$

the sequence of its differences; and

$$\Delta'_0 = \Delta_1 - \Delta_0, \Delta'_1 = \Delta_2 - \Delta_1, \dots\tag{3}$$

the sequence of its second differences, and so on.

It is clear that the sequence (1) is completely determined when its first term and the sequence of its first differences are given; for

$$u_1 = u_0 + \Delta_0, \quad u_2 = u_0 + \Delta_0 + \Delta_1, \quad \dots,$$

$$u_m = u_0 + \Delta_0 + \Delta_1 + \dots + \Delta_{m-1}.$$

Likewise the sequence (1) is completely determined when the first two terms and the sequence of its second differences are given, and so forth. The sequence (1) is called an arithmetic series of n th order when the sequence of its n th differences is constant, and hence the sequence of the $(n+1)$ st differences consists entirely of zeros.

One obtains an arithmetic series of n th order by substituting the numbers $0, 1, 2, 3, \dots$ for x in a whole function $f(x)$ of degree n . For if one puts

$$\Delta_x = f(x+1) - f(x), \quad \Delta_x^{(n-1)} = \Delta_{x+1}^{(n-2)} - \Delta_x^{(n-2)},$$

then Δ_x is of degree $n-1$, Δ'_x of degree $n-2$, and so on with respect to x , and therefore $\Delta_x^{(n-1)}$ is constant.

If now

$$u_0, u_1, u_2, \dots$$

is an arithmetic series of n th order, then the whole series is completely determined when the n values $u_0, u_1, u_2, \dots, u_{n-1}$ and, besides these, the constant n th difference are given. But this latter is determined if also the $(n+1)$ st term u_n is given. We can therefore state the theorem:

An arithmetic series of n th order is completely determined when its first $n+1$ terms are given.

Since a function $f(x)$ of degree n is likewise completely determined by the arbitrarily given values

$$f(0), f(1), f(2), \dots, f(n),$$

it follows that from the whole rational functions of degree n all arithmetic series of n th order are produced when in them the series of natural numbers is substituted for x .

The expression for the general term is then given by formula (7) of the preceding paragraph.

The sums of the first $m+1$ terms of an arithmetic series of n th order,

$$s_m = u_0 + u_1 + \dots + u_m,$$

form an arithmetic series of $(n+1)$ st order, since their first differences

$$s_{m+1} - s_m = u_{m+1}$$

form an arithmetic series of n th order.

Thus, with the help of formula (7) of § 10, the sum s_m can be determined in general when $s_0, s_1, \dots, s_{n+1}$ are assumed known.

To find the generating function $F(x)$ of s_m , when $f(x)$ is the generating function of u_m , put

$$F(0) = f(0), \quad F(1) = f(0) + f(1), \quad F(2) = f(0) + f(1) + f(2), \dots$$

and in formula (7), § 10,

$$F(x) = F(0) + D_0 B_1^{(x)} + D'_0 B_2^{(x)} + \dots$$

one has to put

$$\begin{aligned} D_0 &= F(1) - F(0) = f(1), & D'_0 &= f(2) - f(1) = \Delta_1, \\ D_1 &= F(2) - F(1) = f(2), & D'_1 &= f(3) - f(2) = \Delta_2, \\ D_2 &= F(3) - F(2) = f(3), & D''_0 &= \Delta_2 - \Delta_1 = \Delta'_1, \end{aligned}$$

and so on. Thus one obtains

$$F(x) = f(0) + f(1)B_1^{(x)} + \Delta_1 B_2^{(x)} + \Delta'_1 B_3^{(x)} + \dots \quad (4)$$

For example, take $f(x) = x^2$; then $F(m)$ gives the sum of the first m square numbers. Here

$$\Delta_x = 2x + 1, \quad \Delta'_x = 2,$$

and therefore

$$F(x) = x + 3 \frac{x(x-1)}{1 \cdot 2} + 2 \frac{x(x-1)(x-2)}{1 \cdot 2 \cdot 3} = \frac{x(x+1)(2x+1)}{6}.$$

For $f(x) = x^3$ the same calculation gives

$$F(x) = \left(\frac{x(x+1)}{2} \right)^2.$$

Thus the sum of the first m cubes is equal to the square of the m th triangular number.

§ 12. The polynomial theorem.

In § 8 the following form of the binomial theorem was derived:

$$(x + y)^n = \Pi(n) \sum^{\alpha, \beta} \frac{x^\alpha y^\beta}{\Pi(\alpha)\Pi(\beta)}, \quad (1)$$

where the sum extends over all combinations of two numbers α, β , neither of which is negative, and which satisfy the condition

$$\alpha + \beta = n. \quad (2)$$

This form permits, first by induction, a generalization to the n th power of a polynomial:

$$(x + y + z + \dots)^n = \Pi(n) \sum^{\alpha, \beta, \gamma, \dots} \frac{x^\alpha y^\beta z^\gamma \dots}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma) \dots}, \quad (3)$$

with the determination that $\alpha, \beta, \gamma, \dots$ run through all positive or vanishing integral values which satisfy

$$\alpha + \beta + \gamma + \dots = n. \quad (4)$$

But to prove the correctness of this formula in general, we assume that it is proved when the polynomial contains one term fewer, as is indeed the case when the polynomial contains only two terms.

We then put

$$u = y + z + \dots \quad (5)$$

and apply formula (1) to $(x + u)$, from which there follows

$$(x + y + z + \dots)^n = \Pi(n) \sum^{\alpha, \nu} \frac{x^\alpha u^\nu}{\Pi(\alpha)\Pi(\nu)}, \quad (6)$$

with the restriction

$$\alpha + \nu = n. \quad (7)$$

By the assumption, however, it has already been proved that

$$u^\nu = \Pi(\nu) \sum^{\beta, \gamma, \dots} \frac{y^\beta z^\gamma \dots}{\Pi(\beta)\Pi(\gamma) \dots}, \quad (8)$$

where

$$\beta + \gamma + \dots = \nu. \quad (9)$$

When this is inserted in (6), formula (3) is obtained immediately, and (7) passes over into (4).

The coefficients

$$P_{\alpha, \beta, \gamma, \dots}^{(n)} = \frac{\Pi(n)}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma) \dots}, \quad (10)$$

which, according to their meaning, are whole numbers, are called the polynomial coefficients.

For example, for the third power of the trinomial one obtains

$$\begin{aligned} (x + y + z)^3 &= x^3 + y^3 + z^3 + 3x^2y + 3xy^2 + 3x^2z + 3xz^2 \\ &\quad + 3y^2z + 3yz^2 + 6xyz. \end{aligned} \quad (11)$$